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COURSE CODE: CSA0904
Programming in Java

CO1- ASSESSMENT TOOL - 1

1) Student Management System

Class Student

Private: name, rollno, marks

Method accept()

Method average() = marks / 5

Method result()

IF average $\geq$ 50 print "Pass"

ELSE Print "Fail"

Main

Create Student s

s.accept()

s.result()

2) Library Book Management

Class Book

Private: title, author, ISBN, isavailable

Method issue() $\rightarrow$ isavailable = false

Method returnBook() $\rightarrow$ isavailable = true

Method display()

Main

Create b₁, b₂, b₃

b₁.issue()

b₁.display()

b₁.returnBook()

3) Employee salary calculation

class Employee

Private: empId, name, basicPay, designation

Method gross() = basicPay + HRA + DA

Method net() = gross - PF - Tax

Method payslip()

Main

create Employee e

e.payslip()

4) Shape Area Calculator

class Shape

calculateArea()

class Circle extends Shape

override calculateArea()

class Rectangle extends Shape

override calculateArea()

class Triangle extends Shape

override calculateArea()

Main

call calculateArea() for all Objects

5) Vehicle Registration System

class Vehicle

regno, owner, fuelType

Method display()

class TwoWheeler extends Vehicle

engineCapacity

override display()

class FourWheeler extends Vehicle

seating, AC

override display()

6) Bank Account Hierarchy

class BankAccount

acno, balance, holder

Method withdraw()

class SavingsAccount extends BankAccount

override withdraw()

class CurrentAccount extends BankAccount

override withdraw()

Main

Test withdraw() for both accounts

7) Hospital Patient Record System

class Patient

Private: diagnosis, prescription

Method generateBill()

class Inpatient extends Patient

ward no, charges

override generateBill()

class Outpatient extends Patient

fee
override generateBill()

8) E-commerce product catalog

class product

 applydiscount()

class Electronics extends product

 override applydiscount()

class clothing extends product

 override applydiscount()

class grocery extends product

 override applydiscount()

Main → call applydiscount() for all products

9) University staff Hierarchy

class staff

 calculate Allowance()

class Teaching staff extends staff

class Professor extends Teaching staff

class Lecturer extends Teaching staff

class NonTeaching staff extends staff

Main → call calculate Allowance()

10) Smart Home device controller

class smartDevice

 executeCommand(cmd)

class smartLight extends smartDevice

class smartThermostat extends smartDevice

class smartLock extends smartDevice

Main → Execute "on", "off", "status" for all devices